

COMPUTER VISION · CO1 · TECHNICAL PRESENTATION

Under-Sampling in Digital Images

Defining the Problem, Simulating the Effects, Engineering the Fix

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Why Sampling Rate Is a Design Decision

Every digital image is built by measuring a continuous scene at discrete spatial points. How closely those points are spaced determines whether the image faithfully represents the scene — or silently distorts it.

Medical Imaging

Under-sampled scans can hide or fabricate fine anatomical detail.

Satellite & Aerial

Coarse sampling merges distinct ground features into one pixel.

Video Streaming

Wheels appear to spin backward — temporal under-sampling (stroboscopic effect).

Barcode / Text Scanning

Fine print and thin lines vanish or merge when sampled too coarsely.

Sampling and the Nyquist Criterion

The Sampling Process

A continuous image signal is measured at a finite grid of points (pixels). **The spacing of that grid is the sampling rate.**

Nyquist Criterion

Sampling rate must be at least twice the highest spatial frequency present in the scene, or information is lost irrecoverably.

$$f_s \geq 2 \times f_{ax}$$

f_s = sampling frequency | f_{ax} = highest scene frequency

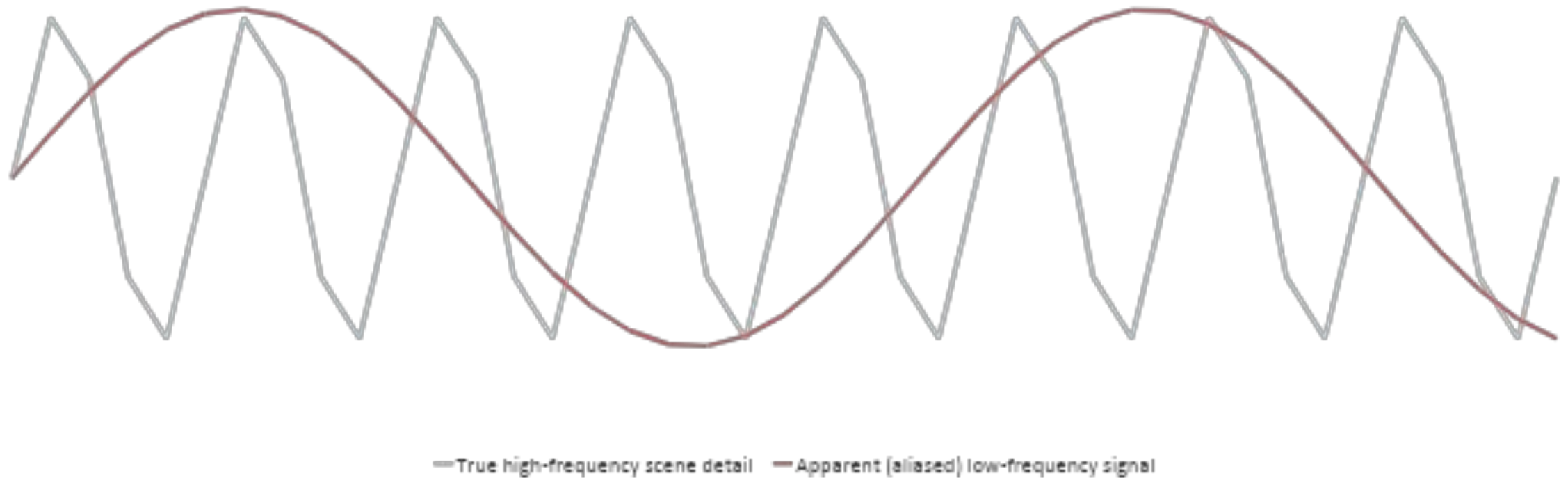
Under-Sampling, Defined

When the sampling rate falls below the Nyquist rate, high-frequency scene detail cannot be represented correctly.

Instead of disappearing, that missed detail reappears disguised as a false, lower frequency — an effect called aliasing.

Key insight: aliasing is not random noise — it is a deterministic, predictable consequence of violating Nyquist.

Aliasing: A Signal in Disguise



The pale line is the true, fine detail in the scene. Sampling too coarsely (undersampling) can only capture a handful of points from it — and those points are equally consistent with the smooth red curve. The reconstructed image displays the wrong, lower-frequency pattern: an alias.

What Under-Sampling Looks Like in Practice

Moiré Patterns

Fine repetitive textures (fabric, screens, fences) produce false wavy interference patterns.

Jagged Edges

Diagonal and curved edges break into staircase artifacts instead of smooth lines.

Loss of Fine Text/Detail

Small fonts or thin structures blur into unreadable smears or vanish entirely.

Temporal Aliasing

Fast-moving objects in video appear to move backward or judder (wagon-wheel effect).

Mitigating Under-Sampling

1

Anti-Aliasing (Low-Pass) Filter

Optically or digitally blur the scene before sampling, removing frequencies above the Nyquist limit.

2

Increase Sensor Resolution

Raise the physical sampling rate (more pixels) so higher scene frequencies satisfy Nyquist.

3

Oversampling + Downsampling

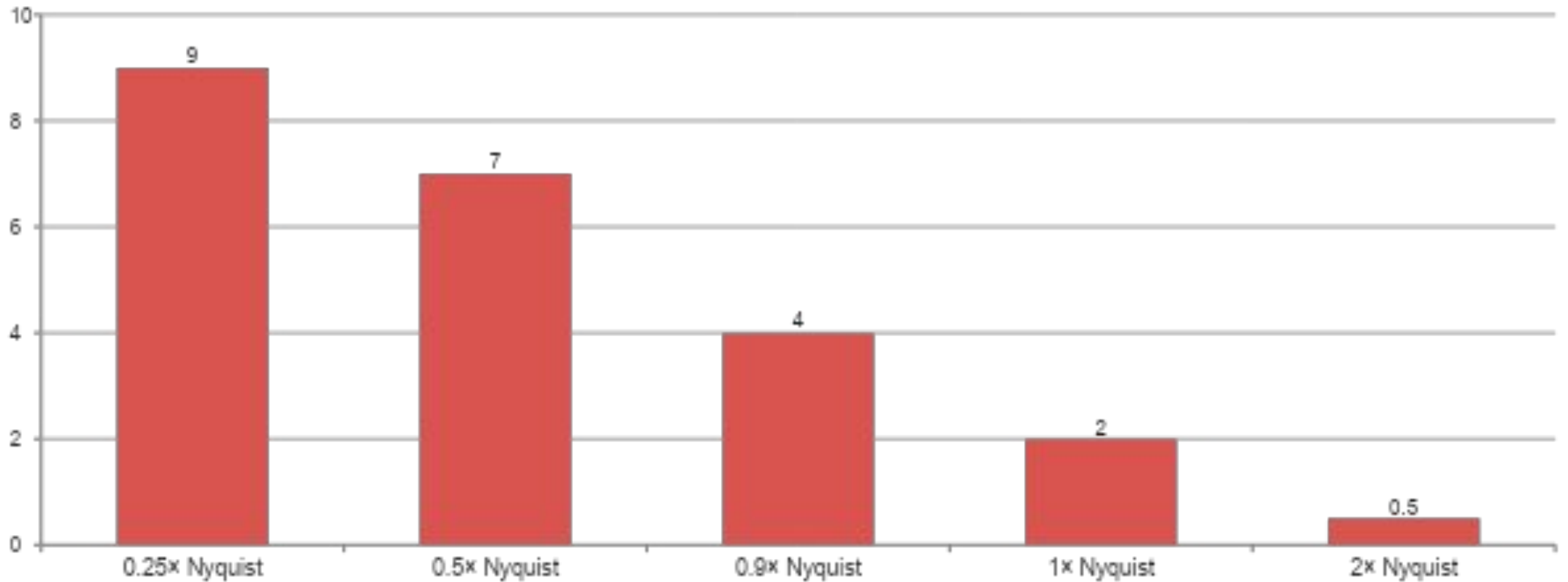
Capture at a higher rate than needed, then filter and reduce — common in scanners and displays.

4

Adaptive / Content-Aware Sampling

Allocate higher sampling density only where scene detail demands it.

Sampling Rate vs Artifact Severity



Artifact severity does not fall off gradually — it drops sharply as the sampling rate approaches and crosses the Nyquist threshold, then stays low beyond it.

Conclusion

- 1 Under-sampling is a violation of the Nyquist criterion, not random noise — it is predictable and preventable.
- 2 It manifests as moiré patterns, jagged edges, lost fine detail, and temporal artifacts like the wagon-wheel effect.
- 3 Anti-aliasing filters and adequate sensor resolution are the primary engineering defenses.
- 4 Every imaging system designer must balance sampling rate against cost, storage, and processing budget.